Choupo

The Open Source Chemical Process Simulator

Explorer Guide

Choupo-dev

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Typeset in LaTeX.

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1 What the Explorer is

The Explorer is Choupo's *interactive thermodynamics bench*: pick components, pick a view, and SEE what the selected property models do — instantly, in the browser, with no case authoring at all. It answers the question that comes *before* any simulation: **"what does this thermodynamics look like?"**

It sits deliberately between two other surfaces:

- The **props bench** (`choupoProps` cases) asks the same questions *reproducibly*: a case file, golden-locked numbers, validation blocks. The Explorer is its sketchpad — what you find here, you pin down there.
- The **simulators** consume the models the Explorer displays. A surprise in a flowsheet (an azeotrope, a scaling mineral, a runaway γ) is usually visible in the Explorer in thirty seconds.

Everything the Explorer draws is computed by the SAME C++ engine (compiled to WebAssembly) with the SAME curated catalogue (`data/standards/`) the solvers use. There is no "display model": if the Explorer shows an azeotrope at $x = 0.89$, the distillation column will hit it at $x = 0.89$.

2 The workflow

1. **Pick components** in the left panel. The list is the live catalogue; each card shows the data the component actually carries (criticals, P^{sat}, c_p, formation — and what it lacks). Case-local components (from an open case's `constant/`) appear alongside, marked.
2. **Pick a view** (the tabs enable themselves when their component-count and data requirements are met; a disabled tab tells you WHY it is disabled).
3. **Pick the method** where more than one applies — activity model, equation of state, P^{sat} correlation. Overlaying two methods on the same axes is the fastest modelling lesson Choupo offers: the gap between the curves IS the modelling uncertainty.
4. **Export**: every view is a Plotly figure (zoom, hover, save PNG) backed by numbers you can carry into a props case for the golden-locked version.

3 The views, one by one

3.1 The compound browser (the Explorer's landing)

The catalogue is a *tree*, not a list. Groups derived from each record's own facts fold and unfold with their member count; the groups a student charges to a vessel open by default, the combustion library, the radicals and the synthetic test stand-ins start folded. A group too large for one screen is subdivided into *families*, and the family comes from the record, never from the name: first the *declared UNIFAC groups* (the highest-precedence group present decides — an acid before an ester before an alcohol, down to an alkane); only for a record with no group decomposition, the *elemental formula*, coarser and labelled so with a "(by formula)" suffix. A record with neither is shown under *"No formula declared in the record"* — a visible curation gap, never a guess. Typing in the search box flattens the tree; a double click inspects a component without changing the set. The browser reads the catalogue; it does not edit a record.

The browser is the left half of the Explorer's *landing* — the screen `?workspace=explore` opens; its right half shows the record of whichever compound you last touched — identity, what the record carries, the curation dossier's verdict — and its **Explore properties** button opens the property surfaces in a tab of their own, carrying the set in the address (`?workspace=properties&components=water,ethanol`), which makes an exploration a link you can hand to somebody. Neither tab needs the other: the surfaces have their own door on the hub, keep a compound rail, and open perfectly well with nothing selected, and a component the address names but the catalogue cannot resolve is left out of the set and said so, never dropped in silence.

3.2 Property vs T/P (scan)

Any pure-component property — P^{sat}, liquid density, c_p, viscosity, enthalpy — swept over temperature or pressure, for one or many components. Use it to: check a correlation's range before trusting it in a case; compare components; spot the nonsense that extrapolation produces outside a fit's window (the guide's recurring lesson: a correlation is DATA, and data has a domain).

3.3 Pure phase diagram (P–T)

The full phase map of one substance: sublimation, melting and vapour-pressure lines meeting at the triple point, ending at the critical point. Expect water's negatively-sloped melting line (the anomaly), and the supercritical region where the liquid/vapour distinction dissolves.

3.4 Binary boiling envelope (T–x–y)

The bubble and dew curves of a binary at fixed P. This is where azeotropes announce themselves: the curves touch, and no distillation can cross the touching point. Switch the activity model (ideal → NRTL → UNIFAC) and watch the envelope bend — ideal solutions cannot azeotrope; real ones do.

3.5 $\gamma(x)$ — activity coefficients

The raw non-ideality of a binary: γ_1, γ_2 against composition, with the infinite-dilution values γ^∞ at the edges. $\gamma > 1$: the pair dislikes mixing (positive deviation, minimum-boiling azeotropes); $\gamma < 1$: association (maximum-boiling). The Gibbs–Duhem constraint ties the two curves together — one cannot bend without the other answering.

3.6 Binary flash (x–y + lever rule)

One flash, drawn: feed, vapour and liquid on the equilibrium diagram with the lever rule made visible. Use it to understand what a single equilibrium stage can and cannot do.

3.7 Binary and ternary LLE

Liquid–liquid splitting from the mixing Gibbs energy: $g_{\text{mix}}(x)$ with the common-tangent construction (binary), and the ternary solubility envelope with tie-lines (UNIFAC-predicted). Expect the tangent points — not the minima — to define the coexisting phases; that distinction is the whole thermodynamics of demixing.

3.8 Ternary boiling surface

T_{bubble} over the composition triangle: valleys are distillation boundaries, and azeotropes sit at the stationary points.

3.9 Scaling (SI vs recovery)

The membrane/brine screening view: concentrate a water composition over recovery and watch each mineral's saturation index climb toward $\text{SI} = 0$ (the precipitation line). The Davies-vs-Pitzer method switch is the lesson: at high ionic strength the two activity models give DIFFERENT scaling verdicts, and choosing one is an engineering decision.

3.10 Equilibrium map (Gibbs)

The chemical-reaction view: pick two or more gas-phase species (with plain formulas — the atom matrix is derived from them, glass-box) and a feed, and get iso-lines of a product's equilibrium mole fraction over the $T \times P$ plane. The whole point is the tension it makes visible: for ammonia ($N_2 + 3H_2 \rightleftharpoons 2NH_3$) the equilibrium optimum sits cold and high-pressure, yet the labelled industrial window sits hot — where the catalyst can work. Answers the beginner's real confusion first ("where do I write the reaction?" — you don't; Gibbs redistributes the ATOMS you fed, shown as a live element balance). *Click any cell* for its full equilibrium composition and the ready-to-run `gibbsReactor` case dict — the map is a case generator. The *advanced* toggle exposes the temperature-approach ΔT (the empirical closeness-to-equilibrium knob commercial tools hide); it snaps rather than animates, and ghosts the $\Delta T = 0$ contours underneath so you read the before/after, not a motion. Unconverged cells are marked, never interpolated.

3.11 Steam tables (IF97)

Saturation and superheat properties of water from IAPWS-IF97 — h_f, h_g, h_{fg}, entropies, volumes — the backbone of every steam and power-cycle case, browsable.

3.12 EduTools (the Methods workspace)

Since 2026-08-15 the classical *method constructions* live in their own workspace, **EduTools**, split from the Explorer by one criterion: *a property surface answers "what does this system's property look like?"* (the Explorer); *a method construction answers "what does the classical graphical method say about a design question?"* — McCabe–Thiele, Kremser, Rayleigh, ε-NTU, pinch composites, and derivation pages for the property models. Open it from the hub's **EduTools** button: it is a tab of its own, needs no case, and does not talk to the flowsheet. A tool is chosen from the top row of the tab; every tool carries a one-line "teaches" caption and its Theory Guide link. Every curve is an engine run (`choupoProps` in WebAssembly) over a transient synthesized case — *zero physics in TypeScript*; the staircase drawn on it is geometry. Planned tools stay visible and disabled, naming what will feed them. It never writes a case: "Author → copy propsDict" emits the dict for you to author on disk.

3.13 Literature ("who measured this?")

Provenance is the differentiator, and this workspace puts its question one click from the flowsheet. With a case open, the tab's menu carries **Literature**: type compound names (every word must match a compound in the article) or click a chip for one of the case's own components, and read who measured that system — authors, year, title, journal, a DOI link, and the properties measured. It searches your *private* mirror of the NIST/TRC ThermoML Archive, installed once with `bin/choupo-thermoml sync` on your own machine; the runtime never reads it and the public site has no mirror, so there it shows the install instructions instead. That absence is a status, not an empty result: *"nobody measured this"* and *"you have no mirror"* are different sentences. It is a *reading list, never a value*: no number is copied into a case or a record, and the only outbound action is opening the article. Read it and cite it — never this index.

4 Provenance and honesty rules

The Explorer inherits the project's credo:

- **Every number has a source.** Component cards show where each value came from; estimates carry their UNVALIDATED badge; nothing is silently filled in.
- **Disabled means explained.** A view that cannot run tells you which requirement failed (component count, missing VLE data, missing UNIFAC groups) — never a dead button.
- **The Explorer never writes.** It reads the catalogue and the open case; promoting data, fitting parameters and locking goldens belong to the props bench and the curation tools. Sketch here, commit there.

5 From Explorer to case: the hand-off

A typical session: you notice in the T–x–y view that your solvent pair azeotropes at 78 % — so the simple column in your flowsheet cannot deliver 99 % purity. The hand-off is: (1) reproduce the envelope as a props case (`propertyScanBinary`) so the finding is golden-locked; (2) switch the flowsheet to the azeotrope-aware design (pressure-swing or entrainer); (3) cite the Explorer finding in the case header. The Explorer is where you LEARN; cases are where knowledge is kept.